

Mathematical inference

science and logic

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*“I got bored one day and I put everything on a bagel...
everything: All my hopes and dreams, my old report
cards, every breed of dog, every last personal ad on
Craigslist, sesame, poppy seed, salt... and it collapsed
in on itself. ’Cause you see, when you really put
everything on a bagel, it becomes this: The truth.”*

Jobu Topaki in Everything Everywhere All At Once

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CHAPTER 1

Prelude: Andante

“Science, including philosophy, is intrinsically polemical, and the polemical cannot be separated from the agonistic. In epochs in which great new things arise, the agonistic factor usually comes to the fore”

Johan Huizinga in *Homo Ludens*

1.1 Inspiration

When I was an undergraduate studying physics, probability theory often felt oddly under-defined. In statistical physics, and even more sharply in quantum physics, probability seemed to appear sometimes as ignorance, sometimes as frequency, sometimes as a physical tendency, and sometimes merely as a useful rule for calculation. The mathematical formalism worked; that was not the problem. The problem was that the interpretations surrounding it seemed to pull in different directions, while the underlying logical story remained incoherent.

After my studies I discovered E. T. Jaynes' *Probability Theory: The Logic of Science* [Jay03]. That book was, for me, the book I would have liked to have read much earlier. Jaynes, also trained as a physicist, refused to treat probability as a bag of unrelated conventions. He presented it instead as a disciplined extension of logic: a way of saying what one should believe, given what one already knows. The attraction of this view was not only technical: It also restored a kind of intellectual honesty to the subject, since hidden assumptions and convenient interpretations could no longer be smuggled in without being noticed.

Yet, Jaynes' book stops before the story is finished. Part of this is literal: the book was posthumously released on the basis of an unfinished manuscript; part of it is conceptual: Jaynes gave a powerful solution for probability, following and extending Cox's programme [Cox46; Cox61], but several of his broader complaints about mathematics, subjectivity, causality and scientific practice remain only partially resolved. This thesis is therefore both a tribute and an attempt at continuation. It asks what Jaynes' programme might look like if it were reconstructed in a constructive, type-theoretic and causal setting.

1.2 Introduction

The central concern of this work is the process by which new knowledge is added to existing information. I will call this process mathematical inference. It is not a marginal technicality; it is, or at the very least should be, the core of scientific reasoning. Jaynes' polemical tone makes sense in this light, and it also fits the opening quotation from Huizinga, where science is described as an intrinsically polemical and agonistic activity [Hui49]. Jaynes was not merely dissatisfied with some specific statistical formulae: He thought that large parts of mathematical and statistical practice were built on foundations that were too permissive or too easily confused with gamesmanship.

Three concerns run through this thesis. First, mathematics should have foundations that make assumptions explicit and prevent uncontrolled use of ideal objects like infinity. Secondly, mathematical inference should be treated as a principled extension of logic, while also being honest about the limits of that derivation. Thirdly, scientific practice should be organised in such a way that good inference is encouraged rather than punished. Jaynes addresses all three, but solves only part of the second. The present work takes these concerns together, since separating them too neatly would already hide part of the problem.

The formal answer developed here starts from Martin-Löf-style type theory

[DP24; Mar84; NPS90]. This is not chosen for decorative foundational purity: It is chosen because constructive type theory forces one to say what is being constructed, in which context, and by which rules. That makes it a natural environment for rebuilding finite probability theory, especially in light of finitary refinements of Cox-style arguments [CW17]. It also connects mathematics to computation, which is important for a thesis in artificial intelligence.

The probabilistic part is not the whole story: Jaynes presents probability theory as a universal logic of science, but the mature formal theory of causal models was developed largely outside his work. Pearl’s structural causal models and do-calculus therefore have to be incorporated into the framework [Pea09; PM18]. Moreover, if probability is called subjective, then subjectivity should not merely mean that one may freely choose a prior. A subjective theory should also be able to represent which questions are in scope, which facts are remembered, which possibilities are being imagined, and how an epistemic state changes. For this reason the later formal development uses contexts, modes and modalities, drawing especially on modal dependent type theory, which extends Martin-Löf-style type theory so that these epistemic concepts can be described natively [Gra+21; Gra23; GSB19].

Finally, inference is not performed in a vacuum. McElreath’s presentation of Bayesian modelling is valuable not only because it teaches calculations, but also because it keeps statistical inference close to modelling choices, scientific questions and workflow [McE20]. The same point becomes sharper at the institutional level: when independent analysts can reach divergent conclusions from the same data, and when large replication projects expose fragile findings, even technically competent inference can become unreliable inside a bad scientific pipeline [Bot+20; Ioa05; Mun+17; Ope15; Sil+18]. Good science needs good inference, but also a good environment in which inference takes place.

Scope and assumptions

This is a thesis in artificial intelligence, where artificial intelligence is understood broadly as the formalisation and implementation of inference. In this sense probability theory, logic, causality and proof checking all fall naturally within its scope. The aim is not to build a large applied AI system, but to investigate what kind of mathematical inference such systems should ideally perform when they reason under uncertainty, act under causal assumptions, or revise an epistemic state.

The thesis therefore looks at inference on two levels. Internally, it asks which formal rules should ideally govern uncertain, subjective and causal inference. Externally, it asks in what scientific environment mathematical and statistical inferences are generally made. This distinction matters, because the proposed formal machinery may not fit comfortably inside every existing scientific habit; and if that turns out to be the case, the problem may lie at least partly with those habits.

The mathematical scope is deliberately restricted: probability spaces are treated as finite or discrete; causal models are treated as finite or discrete; contexts and epistemic states are assumed to be explicitly representable; central objects should, whenever possible, be constructible and implementable. This restriction is not merely a convenience: it follows Jaynes’ suspicion of

1. Prelude: Andante

uncontrolled infinities; it follows the constructive spirit of Martin-Löf-style type theory; and it fits the computational perspective of artificial intelligence.

Selected formal parts will be implemented in Lean 4 where this is useful and feasible. Lean is used here as a proof checker for representative definitions, proofs and examples, not as a promise that every philosophical claim in the thesis will be mechanised: The point is more modest and more important. The central mathematical constructions should be expressible in a form precise enough for independent mechanical checking. Recent discussions of digitised proofs and AI-assisted mathematics make clear that this is becoming an increasingly relevant part of mathematical practice [Kak26; Slo26].

Several limitations follow directly from this scope: this work does not reconstruct all of classical probability theory; it does not provide a general theory of continuous probability spaces; it does not attempt to solve measure theory; it does not claim that Martin-Löf-style type theory is the only acceptable foundation for mathematics; it does not claim that Pearl’s framework captures everything that can meaningfully be called causality; and it does not pretend to give a complete empirical sociology of science. The aim is narrower: to construct a coherent finite framework in which probability, subjectivity and causality can be represented with as little hand-waving as possible.

Research questions

The thesis is guided by the following research questions:

1. Can Jaynes’ probabilistic framework be cast in the language of Martin-Löf-style type theory?
2. Can Jaynes’ probabilistic framework be extended so as to include subjectivity as part of the formal machinery, rather than merely as an interpretation added afterwards?
3. Can Jaynes’ probabilistic framework be extended so as to include Pearl-style causal reasoning?
4. Can Jaynes’ commentary on bad scientific practice be placed in a wider institutional context?

Expected contributions

The first expected contribution is a clarification of Jaynes’ programme. Jaynes’ probability theory will be presented as an attempt to treat probability as extended logic. At the same time, the thesis will separate what Jaynes actually derives from what is claimed. This is important because some of Jaynes’ strongest claims concern universality, subjectivity and completeness, precisely where later objections become sharpest [Col04; CW17; Hal99a; Hal99b; Sha04; Van03].

The second contribution is a constructive reconstruction of the relevant part of probability theory in a Martin-Löf-style type-theoretic setting. In the finite setting, probability assignments should be represented as explicit objects, and the familiar rules of probability should either be derived or obeyed by

construction. The thesis will therefore use probability types where appropriate and connect this to finitary versions of Cox-style reasoning [CW17].

The third contribution is a constructive account of causality. Pearl-style structural causal models will not be replaced, but embedded in a broader type-theoretic and epistemic setting. Interventions, representative counterfactuals and changes of causal structure will be treated as changes between explicitly represented epistemic states.

The fourth contribution is a more explicit account of subjectivity. Subjectivity will not be reduced to the choice of a prior. Instead, epistemic position will be encoded through contexts, modes and transitions between such states. This should make it possible to speak formally about learning, forgetting, interest and hypothesising. This is especially relevant for artificial intelligence, where agents may have bounded resources, limited memory and different bodies of available information [HP07; NR19; Sim55].

The fifth contribution is a limited Lean 4 formalisation. This is not a decorative appendix, but a guard against hidden assumptions. If a definition or proof cannot be made precise, even in the finite setting, it is not an implementation nuisance, but mathematical information.

The final contribution is a diagnosis of scientific practice that connects back to the formal work. Hidden assumptions in mathematics and hidden incentives in science are different phenomena, but they are intertwined. Both make it possible for conclusions to look cleaner than the process that produced them. The thesis therefore ends with practical suggestions concerning policies, incentives and research culture.

Compactly put: the formal part succeeds if finite Jaynes-style probabilistic updating and finite Pearl-style causal reasoning can be represented in the chosen Martin-Löf-style framework, with representative constructions checked in Lean 4. The wider thesis succeeds if this formal work can be convincingly connected to the claim that good science needs both good inference and a good environment in which this inference takes place.

Relevance

The relevance for artificial intelligence is direct. If artificial intelligence is, at least in part, the formalisation and implementation of inference, then its reliability depends on the quality of the reasoning methods on which it is built. A finite, constructive and computational account of probability and causality would provide a clearer route from abstract reasoning to executable systems.

The causal extension is especially important. Modern intelligent systems are increasingly expected not only to predict, but to answer questions about intervention, responsibility and alternative outcomes. Prediction alone is too weak for that task. Pearl’s causal framework shows why prediction has to be supplemented by intervention and counterfactual reasoning [Pea09; PM18]. This thesis asks how such reasoning can be embedded in a more explicitly constructive and subjective framework.

The broader scientific-practice component is also relevant to artificial intelligence. AI systems trained on, used within, or used to accelerate science inherit the weaknesses of the scientific pipeline. If the surrounding environment rewards fragile results, opaque reasoning or publication gamesmanship, then

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faster inference may simply produce bad science more efficiently. The formal and institutional questions therefore have to be kept in view together.

1.3 Outline

The thesis is structured as follows:

The First movement expounds Jaynes' central claims. It discusses his demand for solid foundations in mathematics, solid foundations in statistics, and a good and honest attitude for doing these sciences. It also introduces the mathematical manifesto that will guide the rest of the work.

The Second movement discusses Jaynes' solution for statistics and the problems surrounding it. It reconstructs the Jaynes–Cox view, separates derivation from interpretation, and discusses objections concerning consistency, reasonableness, universality, subjectivity and causality.

The Third movement proposes a more suitable mathematical setting. It discusses alternative logics, type theory, and finally the Martin-Löf-style type theory used in the constructive parts of the thesis.

The Fourth movement uses this mathematical setting to address the problems raised earlier. It develops context manipulation, modalities, probability types and structural causal models inside the chosen type-theoretic framework, and it collects the formal results.

The Fifth movement returns to scientific practice. It expands Jaynes' complaints about bad attitudes into a wider analysis of incentives, institutions and failure modes in modern science, and proposes practical changes.

The appendices contain definitions, proofs, arguments and code-oriented material that would interrupt the main line of the text. They are not less important, but they are placed where they can be checked without interrupting the narrative.

1.4 Idiosyncrasies

Before continuing, a few habits of this text should be made explicit:

The structure is intentionally unconventional. The thesis is arranged as a symphony, with movements rather than ordinary parts. This is inspired by Huizinga's insistence on the agonistic and playful character of culture and science, and by the suspicion that some of this playfulness has been replaced by a more anxious and institutionalised competition. The musical structure is therefore not only decorative: It is a reminder that scientific thought is not merely a sequence of deliverables, but also a playful activity.

Acronyms will be avoided where they do not genuinely help. Science already contains enough abbreviations that sound useful mainly because they sound fundable. Likewise, ordinary terms will not be visually inflated with unnecessary capital letters. This means there will be no grand acronym for “Mathematical Inference as a Logical Extension” (MILE); there will simply be “mathematical inference”, in the sense developed in this text.

The thesis is written with active AI collaboration: ChatGPT has been used for source discovery and general framing; Lumo has been used for stylistic assistance; Claude and ChatGPT are used where appropriate for coding and

proof-checking work, a use that belongs to the wider movement toward AI-assisted formal mathematics [Kak26; Slo26]. This is not hidden, because hiding it would be precisely the wrong lesson for a thesis concerned with explicit assumptions. The intellectual responsibility, including responsibility for errors, remains completely mine.

The intended audience is broad, and this is risky. The thesis lies at the crossing of logic, mathematics, artificial intelligence, philosophy of probability, causality and the sociology of science. Readers from any one of these domains may sometimes want more detail exactly where another reader needs less. The compromise adopted here is narrative-first: the main text preserves the flow of ideas, while technical definitions, proofs and code are moved to the appendices whenever they would otherwise break the story.

These idiosyncrasies are not meant as ornament: They are part of the method. A thesis about explicit reasoning, constructive mathematics and scientific honesty should itself try to be explicit about the way it reasons and the choices it makes. The reader is invited to come along.

CHAPTER 2

First movement: Allegro

“Indeed, an insistence that all of mathematics be stated in that language all of the time can place unnecessary burdens on a theory, particularly one intended for application in the real world. It can also degenerate into an affectation, used only linguistically rather than functionally. To give every old, familiar notion a new, impressive name and symbol unknown to Gauss and Cauchy has nothing to do with rigor. It is, more often than not, a form of gamesmanship whose real purpose is to conceal the Mickey Mouse triviality of what is being done. One would blush to state it in plain English”

E. T. Jaynes in *Probability Theory: The Logic of Science*

2. First movement: Allegro

2.1 Solid foundations for mathematics

2.2 Solid foundations for statistics

2.3 A good and honest attitude for doing these sciences

2.4 Mathematical manifesto

CHAPTER 3

Second movement: Andante

“A story is needed before one can set sail. Look at us. For the sake of the story, we are sitting here aboard a water taxi on our way to the Holy Grail, the unknown, and adventure. People think life becomes more enjoyable through stories. That is true enough, but it is not the point. Without stories, it becomes meaningless. Sense consists of sentences. The words that tell the story lay a reassuring relation of cause and effect over random facts and events. People crave plot because a plot reduces the unbearable and unmanageable chaos of the sublunary world to a human scale, and to a chain of initiatives and consequences that a person is able to comprehend. A plot gives a sense of control, provenance and destination, origin and direction”

Ilja Leonard Pfeijffer in *Grand Hotel Europa*

3. Second movement: Andante

3.1 Jaynes' + Cox theory, what is derived

3.2 Objections

Consistency

Reasonableness

Universality

Subjectivity

Causality

CHAPTER 4

Third movement: Scherzo

“A common mistake that people make when trying to design something completely foolproof is to underestimate the ingenuity of complete fools”

Douglas Adams in *Mostly Harmless*

4. Third movement: Scherzo

4.1 Different type of logics

4.2 Type theory

4.3 Martin-Löf-style type theory

CHAPTER 5

Fourth movement: Vivace

*“Scientists have calculated that the chance of anything so
patently absurd actually existing are millions to one.
But magicians have calculated that million-to-one chances crop
up nine times out of ten”*

Terry Pratchett in *Mort*

- 5.1 Context manipulation in Martin-Löf-style type theory**
- 5.2 Modalities inside Martin-Löf-style type theory**
- 5.3 Probability types**
- 5.4 Structural causal models in Martin-Löf-style type theory**
- 5.5 Results**

CHAPTER 6

Fifth movement: Presto

“For far too long economists have sought to define themselves in terms of their supposedly scientific methods. In fact, those methods rely on an immoderate use of mathematical models, which are frequently no more than an excuse for occupying the terrain and masking the vacuity of the content. Too much energy has been and still is being wasted on pure theoretical speculation without a clear specification of the economic facts one is trying to explain or the social and political problems one is trying to resolve”

Thomas Piketty in *Capital in the Twenty-First Century*

Acknowledgements

“Teleutias truly sailed home under the happiest of circumstances. When he went down to the sea to begin his journey home, there was not a soldier who did not clasp his hand. One placed a wreath on his head, another put a fillet around him, and those who came too late, when he was already putting out to sea, still threw wreaths into the water and wished him every good thing.

I realize that I am not describing any great expenditure, any danger, or any strategy worth mentioning; but by Zeus, I do think it is important for a man to consider what kind of conduct enabled Teleutias to evoke such a response from his subordinates, for that is, beyond any amount of money or danger, a man’s most noteworthy achievement”

Xenophon in *Hellenica*

Appendices

APPENDIX A

Definitions

“So if you are an expert and, as I said just now, have deceived me with your promise to give a demonstration about Homer, you are playing a trick on me. But if you are not an expert, but are possessed by Homer through a divine gift and, without knowledge, say many beautiful things about the poet, as I claimed is the case with you, then you are not playing a trick on me. So you need only choose how you wish us to regard you: as a man who plays tricks, or as a man governed by the gods”

Socrates in *Ion*

APPENDIX B

Proofs and arguments

“The important thing is life — life and nothing else! What is any ‘discovery’ whatever compared with the incessant, eternal discovery of life?”

Hippolyte in *Idiot*

B. Proofs and arguments

- B.1 Causality argument**
- B.2 Physics argument**
- B.3 Excluded-middle proof**
- B.4 Clayton proof**
- B.5 SCM-equivalence proof**
- B.6 Causality-completeness proof**

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“Contrarily, it is characteristic of a truly wise man that, being just a mortal, he doesn’t want to transgress any boundaries with his wisdom and together with humanity either turns a blind eye, or, out of courtesy, wanders along”

Folly in *The Praise of Folly*

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